

Supriya Deb

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Education

Techno Main Salt Lake, Kolkata

Aug 2021 – Jul 2025

B.Tech. in Computer Science and Engineering (Data Science)

Work Experience

Data Analyst Intern, Exelux.in (Remote)

Sep 2025 – Nov 2025

- Extracted, cleaned, and validated user behavior data using Python (Pandas) and SQL, ensuring high-quality datasets for reliable analysis across multiple sources.
- Built a K-means customer segmentation model to identify distinct user cohorts, uncovering behavioral patterns and improving targeted analysis of acquisition channels.
- Analyzed segment performance and translated insights into actionable recommendations, supporting data-driven decision-making for marketing optimization.

Projects

Instagram Content Strategy Agent

 [GitHub](#)  [Live](#)

Python, scikit-learn, Django, Groq, Pandas

- Trained and compared **5 regression models** to predict Instagram impressions; $R^2 = 0.9485$ and $CV R^2 = 0.82$ using Ridge Regression, with documented feature engineering and evaluation experiments.
- Developed a custom **analytics scoring engine** in Django to compute viral scores, engagement rates, and multi-post performance forecasts with trend detection and session-history calibration.
- Designed a **rule-based diagnostic layer** to evaluate post performance and generate actionable growth recommendations, persisted using **Django ORM** and SQL-backed session storage.
- Integrated a **Groq LLM agent (Llama 3.3-70B)** for **NLP-driven** conversational analytics, using dynamic system prompts grounded in user post history; implemented a deterministic fallback for reliability without API dependency; **deployed on Render**.

Histopathological Image Segmentation (U-Net + BASNet)

 [GitHub](#)

Python, TensorFlow

- Manual nuclei annotation is error-prone and slow; automated segmentation on the **MoNuSeg dataset** using **TensorFlow** and **U-Net + BASNet**, achieving **Dice: 0.76**, **IoU: 0.78**, and **Boundary F1: 0.82**.
- Limited data risked overfitting; built a preprocessing pipeline with **normalization**, **resizing**, and **data augmentation** to improve generalization across staining variations.
- Standard loss functions missed boundary detail; implemented combined **Dice**, **Focal**, and **SSIM loss** to explicitly penalize boundary errors.
- Stabilized training on a small high-variance dataset using **early stopping** and **hyperparameter tuning**; documented training experiments, loss comparisons, and segmentation results.

Technical Skills

Programming: Python, SQL

Machine Learning: Scikit-learn, TensorFlow, Regression, Random Forest, Gradient Boosting, K-Means

Data Analysis: Pandas, NumPy, EDA, Statistical Analysis

Visualization: Power BI, Excel, Matplotlib

Databases & Cloud: PostgreSQL, SQL Server, Snowflake, AWS (S3, Redshift)

AI: LLMs, Prompt Engineering

CS Fundamentals: OOP, DBMS, Computer Networks, Statistics & Probability

Certificates

- Data Analysis and Visualization with Power BI — Coursera (Microsoft)
- Databases and SQL for Data Science with Python — Coursera (IBM)